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ReGAT-BERT: Transformer-Graph Fusion for Dynamic Reranking

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Abstract

Effective and precise passage retrieval is crucial in contemporary information retrieval systems, particularly in light of the exponential expansion of digital document collections. Conventional approaches relying on lexical matching frequently struggle to capture semantic subtleties, often overlooking passages that are semantically pertinent but lexically divergent from user queries. Recent advancements in transformer-based models, such as BERT and its variants, have substantially enhanced retrieval performance by facilitating a deeper semantic comprehension. Nonetheless, the computational demands of these models present scalability challenges when applied to large datasets.

In this work, we propose an innovative passage retrieval framework that seamlessly integrates lexical-semantic relevance with contextual information, markedly boosting retrieval precision and coherence. Our methodology unfolds in three stages: first, a Cross-Encoder BERT model assesses semantic relevance by analyzing fine-grained interactions between queries and passages. Second, a Bi-Encoder architecture generates initial embeddings, which are subsequently refined using a Graph Attention Network (GAT) to incorporate structural and contextual relationships among passages. Third, a dynamic reranking mechanism improves ranking by merging semantic relevance scores with contextual similarity evaluations, ensuring that retrieved passages are both contextually coherent and semantically precise.

Evaluations conducted on benchmark datasets against state-of-the-art baselines reveal that our hybrid approach enhances retrieval accuracy and coherence. Furthermore, by integrating graph-based embedding refinements, our model mitigates the scalability constraints typically associated with Cross-Encoder architectures.

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1. Introduction

Effective and accurate passage retrieval is a crucial component of contemporary information retrieval systems, significantly influencing user satisfaction and the overall efficiency of accessing information. As digital content con-

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tinues to grow exponentially, the challenge of ensuring that the most relevant passages are prioritized in search results persists. Conventional lexical matching techniques, like BM25 [12], often face difficulties with semantic subtleties, frequently inadequately ranking passages that are semantically relevant but lexically dissimilar. For instance, a query such as "how to treat a common cold" might overlook highly pertinent passages discussing "remedies for flu symptoms" due to differences in wording.

Recent progress in transformer-based architectures, particularly BERT [2], has led to substantial improvements in semantic understanding for retrieval tasks. Cross-Encoder models excel at capturing detailed semantic relationships between queries and passages, thereby boosting initial retrieval accuracy [3]. Nevertheless, despite their strong semantic precision, these models encounter scalability limitations due to their high computational demands, posing challenges for practical deployment. In contrast, Bi-Encoder architectures, such as Sentence-BERT (SBERT) [11], mitigate computational inefficiencies but sacrifice accuracy as queries and passages are encoded separately.

Additionally, passage retrieval systems frequently face challenges with insufficient reranking, causing relevant passages to appear lower in the results, which negatively impacts user experience. Existing approaches often overlook contextual and structural information, which could significantly enhance ranking accuracy. Recent studies using Graph Neural Networks (GNNs), especially Graph Attention Networks (GATs), show promising potential in capturing the contextual and structural relationships between passages, thereby improving retrieval coherence [16, 10]. Nevertheless, purely graph-based methods may fall short in addressing the critical need for immediate semantic alignment, which is vital for accurately understanding user intent.

To address these limitations and significantly enhance reranking performance, we introduce a novel hybrid retrieval approach that integrates transformer-based semantic relevance with GAT-driven contextual modeling. Our framework is structured around three core stages: (i) the Lexical-Semantic Relevance Phase, where a Cross-Encoder meticulously assesses semantic relevance by analyzing detailed semantic interactions; (ii) the Graph-Enhanced Contextual Similarity Phase, which enriches initial Bi-Encoder embeddings using a Graph Attention Network (GAT) to capture broader structural and contextual relationships; and (iii) the Dynamic Reranking Phase, where semantic relevance and contextual similarity scores are dynamically combined to prioritize the most relevant passages. This systematic methodology ensures that retrieved passages are not only semantically aligned but also contextually consistent, reliably placing the most pertinent results at the forefront of retrieval outcomes.

The primary objective is to create a hybrid retrieval framework that seamlessly combines semantic relevance and contextual structure using a dynamic reranking approach, ensuring that the most informative and contextually coherent passages are prioritized during inference.

The remainder of this paper is structured as follows: Section 2 reviews related works. Section 3 details the proposed methodology. Section 4 presents the experimental evaluations, and Section 6 concludes with a summary and outlines future research directions.

2. Related Works

Passage retrieval and ranking are essential components in the fields of information retrieval and natural language processing, playing a pivotal role in shaping user experiences. Over time, research has progressed from simple lexical matching techniques to more sophisticated models that integrate semantic comprehension and structural context. This section provides a concise overview of transformer-based and graph neural network-based methods, highlighting their advantages and drawbacks to frame the context for our proposed hybrid approach.

2.1. Transformer-based Models

Effective passage retrieval and ranking are fundamental tasks in information retrieval and natural language processing research. Traditional ranking methods primarily rely on lexical matching algorithms, such as BM25 [12], which use term frequency and inverse document frequency to rank passages. While efficient, these lexical approaches often struggle to handle semantic nuances, especially paraphrasing and semantic similarity [3].

To overcome these limitations, recent advancements in deep learning have introduced transformer-based architectures, like BERT [2], which have enhanced semantic understanding in retrieval tasks. Ouni et al. [8] proposed a classification approach combining BERT with topic-based features and convolutional neural networks (CNN), illus-

trating that contextual embeddings can be effectively enhanced with handcrafted semantic features. Cross-Encoder architectures jointly process queries and passages, providing substantial performance improvements by capturing fine-grained semantic interactions [1]. This deep contextual modeling has shown strong performance but suffers from high computational costs, limiting their scalability. On the other hand, Bi-Encoder models, such as Sentence-BERT (SBERT) [11], alleviate this issue by generating independent dense embeddings for queries and passages, facilitating efficient similarity computation.

BERT-based CNNs have also been applied to extract semantic-stylistic features for classification and scaled to sentiment analysis using DistilBERT within a distributed framework [9, 4].

2.2. Graph Neural Network-based Models

Although transformer-based models excel in semantic understanding, they typically overlook contextual and structural relationships within documents. The emergence of Graph neural network (GNN) approaches, particularly Graph Attention Networks (GATs) [13], have gained popularity for incorporating structural context into retrieval tasks. These models capture relational structures among passages, significantly enhancing retrieval accuracy [16]. For example, Graph Retrieval [15] explicitly uses graph neural networks to propagate contextual information, effectively modeling intricate structural relationships across documents. The primary advantage of GNN models stems in their ability to incorporate document-level context, which improves coherence and accuracy. However, their reliance on structural information may lead to less sensitivity to direct semantic interactions between query and passage, potentially compromising immediate semantic relevance.

Hybrid models combining transformer-based semantic embeddings and GNN-enhanced contextual information have emerged as promising solutions [7]. Methods such as Graph-Enhanced Transformer (GET) [5] and Contextualized Embeddings with Graph Attention [14] demonstrate how integrating semantic and structural contexts through graph attention mechanisms significantly refines retrieval outcomes.

These studies demonstrate the flexibility of transformer-based models and the added value of integrating structural context through GNNs, confirming their complementary roles in enhancing retrieval effectiveness across diverse scenarios.

3. Proposed ReGAT-BERT Details

Our proposed method, ReGAT-BERT, overcomes the limitations of transformer-based and GAT-based models by seamlessly combining the semantic precision of transformer architectures with the contextual coherence of GAT. This integration ensures a balanced approach, capturing both immediate semantic relevance and structural contextual relationships, thereby significantly improving retrieval performance. ReGAT-BERT is specifically designed to efficiently extract and prioritize the most relevant passages from large document collections, greatly enhancing the ability to address user queries. Our main objective is to advance passage ranking by using both lexical-semantic relevance and contextual relationships, enabling a deeper and more nuanced understanding of user queries and document content.

3.1. Different stages of the proposed approach

The method involves three distinct yet interconnected phases, illustrated in Figure 1.

- First, we introduce the **Lexical-Semantic Relevance** phase, wherein a Cross-Encoder based on BERT meticulously evaluates direct alignment between queries and passages. By modeling fine-grained lexical and semantic interactions, this phase generates precise semantic relevance scores, capturing the immediate textual compatibility between user queries and candidate passages.
- Next, the **Graph-Enhanced Contextual Similarity** phase addresses the intrinsic relationships among passages across the entire document collection. Here, a Bi-Encoder generates initial embeddings, which are subsequently enriched through a Graph Attention Network (GAT). This graph-based refinement process incorporates structural and semantic dependencies, producing sophisticated contextual similarity scores that reflect broader document-level interactions and relationships.

- Finally, the **Dynamic Reranking** phase synthesizes insights from the previous two phases by strategically combining semantic relevance and contextual similarity scores. Through this integrated approach, passages are dynamically reranked, ensuring that the most contextually coherent and semantically pertinent passages are prioritized and delivered first. This effective strategy substantially enhances retrieval effectiveness by simultaneously considering local passage-query alignment and global contextual coherence.

Collectively, our method integrates state-of-the-art (as [section 2](#)) transformer models with graph neural network techniques, offering a robust, interpretable, and contextually informed passage retrieval framework that significantly surpasses traditional ranking strategies.

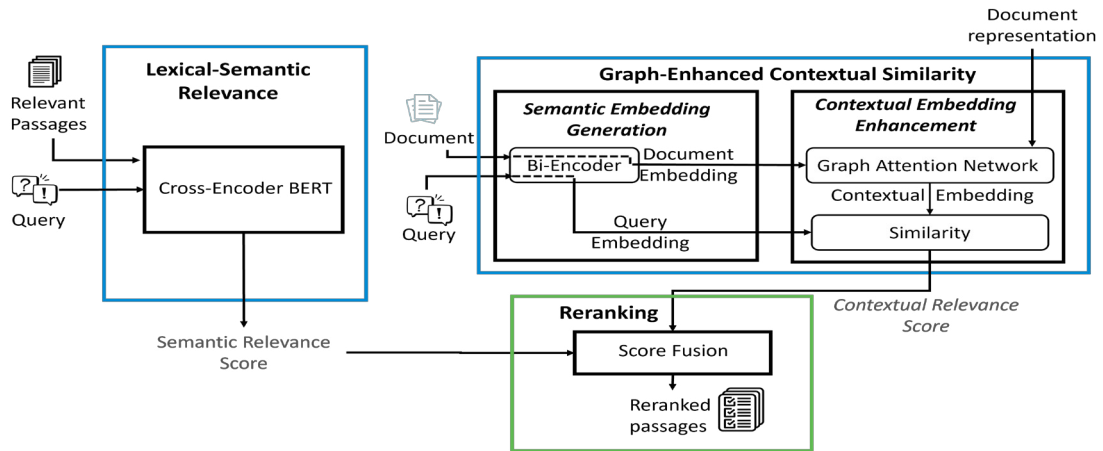


Fig. 1. Overview of the ReGAT-BERT architecture

3.2. Lexical-Semantic Relevance

The main goal of this phase is to accurately assess passage relevance by modeling fine-grained semantic interactions between the query and candidate passages. In our approach, we use a Cross-Encoder based on BERT to refine the evaluation of passage relevance in response to a given query. Unlike Bi-Encoder models, which generate independent representations for queries and passages, the Cross-Encoder processes these elements jointly, enabling a more precise modeling of semantic relationships.

This phase is designed to compute the semantic relevance score, which quantifies the likelihood that a passage adequately answers the query. The model takes as input a query-passage pair, encoding it through BERT's transformation layers to capture contextual relationships between terms. However, a major challenge of the Cross-Encoder lies in its computational cost, as it requires evaluating each query-passage pair individually.

To alleviate this limitation, we apply a filtering mechanism that limits Cross-Encoder evaluation to a smaller group of passages that are more likely to be relevant. This filtering relies on a relevance propagation strategy that estimates the potential importance of passages based on the contextual signals of their parent sections or document structure. The idea is to propagate relevance scores from higher-level nodes (such as section titles or document headings) to child passages, using lexical or concept-based similarity. This significantly reduces the number of passages processed, improving computational efficiency while maintaining retrieval effectiveness.

3.3. Graph-Enhanced Contextual Similarity

The objective of this section is to compute the similarity between a query and a passage by using the query embedding generated by a Bi-Encoder and the contextual passage embedding refined through a Graph Attention Network (GAT). This approach ensures that passage ranking is informed not only by direct semantic relevance but also by structural and relational context within the document collection.

3.3.1. Semantic Embedding Generation

It is a crucial step in our retrieval framework, designed to encode the semantic meaning of both queries and passages. Using dense vector representations, this phase overcomes the limitations of traditional lexical matching methods, enabling a richer understanding of textual content.

In this phase, we use a Bi-Encoder architecture, such as Sentence-BERT (SBERT), to independently encode queries and document passages into dense vector embeddings. Specifically, given a query q and a passage p from a document D_i , the input texts are tokenized with special tokens ([CLS] and [SEP]) to clearly distinguish the boundaries between different text segments. For instance, the query is formatted as "[CLS] query [SEP]" and the passage as "[CLS] passage [SEP]". The Bi-Encoder then produces two independent embeddings as expressed in Equation 1

$$E_q = f_\theta([CLS] \ q \ [SEP]), \quad E_p = f_\theta([CLS] \ p \ [SEP]) \quad (1)$$

where f_θ represents the transformer-based encoding function parameterized by the SBERT model. The embeddings for queries and passages are precalculated independently, enhancing retrieval efficiency.

These semantic embeddings serve as foundational representations for subsequent phases, enabling efficient retrieval.

3.3.2. Contextual Embedding Enhancement

Graph Attention Network. While semantic embeddings effectively capture textual meaning, they typically do not account for the structural and relational context inherent in document collections. The Contextual Embedding Enhancement phase addresses this limitation by refining semantic embeddings with structural and relational information derived from document organization and relationships, enabling a more contextually informed retrieval.

To refine passage embeddings with contextual structure, we employ GATs, a class of multi-layer graph neural networks that enhance node representations through learned attention-based message passing. Unlike traditional Graph Convolutional Networks (GCNs), which aggregate information uniformly from neighboring nodes, GATs dynamically assign attention weights to neighboring nodes, allowing the model to focus more relevant contextual signals. Each GAT layer computes attention-weighted aggregations of node embeddings, effectively enabling each node to selectively integrate relevant contextual information from its neighbors. By stacking multiple GAT layers, the model effectively propagates contextual information from progressively distant neighbors, capturing hierarchical and relational dependencies that influence passage ranking.

Formally, given a graph $G = (V, E)$ each node $i \in V$ represents a passage, initially embedded using the semantic embeddings from the SBERT. Edges $e_{ij} \in E$ reflect structural relationships such as intra-document hierarchies or cross-references between passages. For node i , we define its set of neighbors N_i . Each GAT layer computes updated embeddings h'_i for node i as described in Equation 2:

$$h'_i = \sigma \left(\sum_{j \in N_i \cup \{i\}} \alpha_{ij} W h_j \right) \quad (2)$$

where:

- h_j represent the input embeddings of nodes j .
- W is a learnable weight matrix applied to node embeddings.
- α_{ij} represents attention coefficients, calculated using an attention function specifically defined as shown in Equation 3:

$$\alpha_{ij} = \text{softmax}_j(\text{LeakyReLU}(a^T [W h_i || W h_j])) \quad (3)$$

Here, a is a learned attention vector, h_i represent the input embeddings of nodes i , and the operator $||$ indicates vector concatenation.

Our GAT model further employs a multi-head attention mechanism, computing multiple attention operations in parallel. Each attention head captures different structural and semantic patterns in the data, with their outputs subsequently concatenated or averaged, thus enhancing robustness and expressiveness.

Additionally, our approach identifies relationship types through edge-specific weight matrices W_r , enabling differentiated treatment of structural relationships (e.g., hierarchy versus citation). This ensures nuanced context propagation tailored to the nature of connections.

The embeddings generated by the GAT effectively capture structural dependencies, enriching passage representations with contextual information. These refined passage embeddings are then used in the next step to compute similarity scores, ensuring that passage ranking accounts for both semantic relevance and document structure.

Similarity. Evaluating passage relevance within a document collection requires an effective similarity measure that integrates both the semantic intent of the user’s query and the contextual characteristics of candidate passages. This phase quantifies the alignment between the query embedding E_q , generated using the Bi-Encoder (SBERT), and the contextually enriched passage embedding E_{P_i} , refined through Graph Attention Network (GAT)-based modeling. This integrated similarity measure captures semantic content alongside structural context, ensuring that retrieved passages are highly relevant to the query and coherent within the broader document structure.

Specifically, the similarity $Sim_{ctx}(q, P_i)$ between the query q and the passage P_i is computed using cosine similarity, as formally presented in Equation 4:

$$Sim_{ctx}(q, P_i) = \frac{E_q \cdot E_{P_i}}{\|E_q\| \|E_{P_i}\|} \quad (4)$$

where E_q and E_{P_i} represent, respectively, query embedding and passage embedding. $E_q \cdot E_{P_i}$ represents the scalar product, and $\|E_q\|, \|E_{P_i}\|$ represents the Euclidean norms of the embeddings.

3.4. Reranking

The reranking phase aims to refine the ranking of retrieved passages by integrating semantic relevance scores $Sim_{sim}(q, p)$ (computed using the Cross-Encoder BERT) and contextual relevance scores $Sim_{ctx}(q, p)$ (derived from Graph-Enhanced Contextual Similarity, integrating Bi-Encoder SBERT and GATs). Unlike traditional score aggregation techniques, our approach uses a learning-based optimization strategy to dynamically adjust weighting parameters based on labeled data.

This approach is motivated by the fact that semantic and contextual relevance signals capture complementary aspects of relevance. While semantic similarity reflects the direct alignment between a passage and a query in a dense embedding space, contextual similarity incorporates structural and relational dependencies, capturing how passages fit within the overall document hierarchy. This understanding directly informs how we compute the reranking score for each passage.

The final reranking score $reRankScore_i$ for passage P_i is computed as described in Equation 5:

$$reRankScore_i = \alpha \cdot Sim_{sim}(q, P_i) + \beta \cdot Sim_{ctx}(q, P_i) \quad (5)$$

where $Sim_{sim}(q, P_i)$ and $Sim_{ctx}(q, P_i)$ are, respectively, the semantic and contextual relevance scores between query q and passage P_i . α and β are learnable parameters that determine the contribution of these components to the final reranking.

To optimize these parameters, we define a binary cross-entropy loss function shown in Equation 6:

$$L(\alpha, \beta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(reRankScore_i) + (1 - y_i) \log(1 - reRankScore_i)] \quad (6)$$

where y_i is the ground-truth relevance label (1 for relevant, 0 for non-relevant), and $reRankScore_i$ is the predicted ranking score for passage P_i .

To minimize the loss function, we apply a gradient-based optimization algorithm, iteratively updating the parameters α and β which are formulated in the next Equations 7 and 8:

$$\alpha_{new} = \alpha_{old} - \eta \frac{\partial L}{\partial \alpha} \quad (7)$$

$$\beta_{\text{new}} = \beta_{\text{old}} - \eta \frac{\partial L}{\partial \beta} \quad (8)$$

where η is the learning rate controlling the step size for updates. This adaptive learning ensures that the reranking function dynamically adjusts to different query types and document structures, improving overall retrieval performance.

Algorithm 1 summarizes the reranking procedure. It takes semantic and contextual scores for each passage, computes the final reranking score using a weighted linear combination. The parameters α and β are optimized via gradient descent, and passages are classified into relevance categories to enhance reranking effectiveness.

Algorithm 1: Learning-Based Semantic-Contextual Reranking

Input : $S_{\text{sim}}(P)$: Semantic relevance score from Cross-Encoder BERT

$S_{\text{ctx}}(P)$: Contextual relevance score from SBERT + GAT

α, β : Initial fusion parameters

$y(P)$: Ground-truth relevance labels

η : Learning rate

Output: $\text{reRankScore}(P)$: Computed reranking score

for each passage P_i **do**

$\text{reRankScore}_i = \alpha \cdot S_{\text{sim}}(q, P_i) + \beta \cdot S_{\text{ctx}}(q, P_i)$

for each training iteration do

$L(\alpha, \beta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\text{reRankScore}_i) + (1 - y_i) \log(1 - \text{reRankScore}_i)]$

$\alpha_{\text{new}} = \alpha_{\text{old}} - \eta \frac{\partial L}{\partial \alpha}$

$\beta_{\text{new}} = \beta_{\text{old}} - \eta \frac{\partial L}{\partial \beta}$

return reRankScore_i

4. Experimental study and results analysis

This section presents the evaluation of our ReGAT-BERT model, comparing it with both internal baselines and state-of-the-art retrieval models. The evaluation is conducted on the MS MARCO Passage Ranking dataset using MRR@10 and NDCG@10 as the primary evaluation metrics.

4.1. Experimental design

To evaluate the effectiveness of ReGAT-BERT, we conducted experiments on the MS MARCO Passage Ranking dataset, a large-scale benchmark commonly used for passage retrieval tasks. This dataset consists of real-world user queries paired with relevant and non-relevant passages, making it suitable for assessing retrieval performance in a realistic setting.

4.1.1. Dataset

We evaluate our proposed method using the MS MARCO (Microsoft MACHine Reading COMprehension) dataset, a large-scale benchmark specifically designed for assessing passage retrieval and reranking methods in information retrieval. The dataset comprises 8.8 million passages extracted from web documents and approximately 1 million training queries. Additionally, it includes a development set for evaluation, ensuring that each query is associated with at least one relevant passage.

4.1.2. Baselines

To assess the effectiveness of the proposed ReGAT-BERT model, we conduct a comparative evaluation against both internal components and external state-of-the-art baselines.

First, we compare ReGAT-BERT with its internal submodules used in isolation:

- **CE-BERT (Cross-Encoder BERT):** which performs fine-grained semantic matching between queries and passages without contextual enhancement.
- **SBERT (Sentence-BERT):** a Bi-Encoder model that independently encodes queries and passages into dense vectors, serving as the base embedding model in ReGAT-BERT.

We then extend the comparison to include external baselines that represent various paradigms in passage retrieval:

- **BM25:** A traditional lexical matching model based on term frequency and inverse document frequency.
- **$BERT_{base}$:** A standard transformer model used in a cross-encoder setting for reranking tasks.
- **$BERT_{large}$:** A larger variant of BERT with more parameters, capable of capturing deeper semantic relationships.
- **ColBERT:** A late-interaction model that balances efficient retrieval and token-level matching by decomposing query and passage interactions.
- **DPR (Dense Passage Retriever):** A dual-encoder architecture trained on MS MARCO, enabling scalable dense retrieval.
- **Graph-RoBERTa:** A graph-augmented retrieval model that integrates structural document relationships into transformer-based representations.

This diverse set of baselines allows for a detailed evaluation of the contributions brought by semantic modeling, contextual graph reasoning, and adaptive reranking within the ReGAT-BERT method.

4.2. Results and discussion

This section presents the evaluation results of ReGAT-BERT compared to both internal components and external state-of-the-art models. The evaluation is based on the MS MARCO Passage Ranking benchmark, using MRR@10 and NDCG@10 as the primary metrics. Our goal is to assess the contribution of semantic modeling, contextual graph reasoning, and dynamic reranking.

4.2.1. Internal Comparison

Table 1 presents a comparison of ReGAT-BERT with two internal baselines: CE-BERT and SBERT. While CE-BERT captures fine-grained semantic interactions between queries and passages, SBERT provides a bi-encoder representation of textual similarity without query-passage interaction. The results indicate that ReGAT-BERT outperforms both CE-BERT and SBERT in terms of MRR@10 and NDCG@10, demonstrating the effectiveness of incorporating GAT and dynamic reranking. The increase in NDCG@10 confirms that our model ranks relevant passages higher than competing internal models. The improvement over CE-BERT suggests that integrating contextual relationships between passages enhances retrieval effectiveness. Compared to SBERT, our model achieves a significant gain, confirming that dynamic reranking and graph-based refinement improve passage ranking accuracy.

4.2.2. External Comparison

Table 2 and Figure 2 presents a comparison of ReGAT-BERT with several state-of-the-art passage retrieval models, including traditional and neural approaches. The results indicate that ReGAT-BERT outperforms all the evaluated metrics, underscoring the effectiveness of integrating GAT and dynamic reranking mechanisms.

Compared to $BERT_{large}$, ReGAT-BERT achieves an improvement of 13.2% in MRR@10 and 8.3% in NDCG@10. In comparison with $BERT_{base}$, the gains are even more pronounced, with an increase of 19% in MRR@10 and 12% in NDCG@10. These improvements highlight the advantage of incorporating structural context through graph reasoning and demonstrate the benefits of adaptive reranking over static semantic scoring.

Furthermore, the improvement over ColBERT highlights the advantage of using both document structure and semantic matching beyond fine-grained token-level interactions. The notable performance margin compared to Graph-RoBERTa and DPR further suggests that ReGAT-BERT provides a more robust model for passage ranking, capable of capturing both local semantic precision and broader contextual coherence. In addition, a considerable improvement is observed in Recall@1000, which shows that the model is able to retrieve more relevant passages during the first retrieval stage, making it more effective for large-scale search tasks.

Table 1. Comparison of ReGAT-BERT with Internal Models

	MRR@10	NDCG@10
CE-BERT	0.390	0.743
SBERT	0.330	0.690
ReGAT-BERT	0.413	0.794

Table 2. Comparison of ReGAT-BERT with State-of-the-Art Baselines

	Recall@1000	MRR@10	NDCG@10
BM25 [6]	0.814	0.167	0.228
$BERT_{base}$	-	0.347	0.709
$BERT_{large}$	-	0.365	0.733
ColBERT [6]	0.814	0.349	0.401
DPR [6]	0.952	-	0.177
Graph-RoBERTa	-	-	0.509
ReGAT-BERT	0.978	0.413	0.794

4.2.3. Influence of α on Reranking Performance

Figure 3 illustrates the impact of varying the semantic relevance weight α on the overall performance of the reranking process. The results show that the highest MRR@10 is achieved when $\alpha = 0.6$, while NDCG@10 reaches its peak at $\alpha = 0.65$. These findings suggest that combining semantic and contextual signals leads to optimal performance when the model places slightly more emphasis on semantic alignment, while still incorporating a meaningful degree of contextual information. The sharp decline in performance when α approaches 1.0 (purely semantic) or 0.0 (purely contextual) confirms the complementary nature of both components. This balance, automatically learned during training, allows the model to generalize effectively across different query types and document structures.

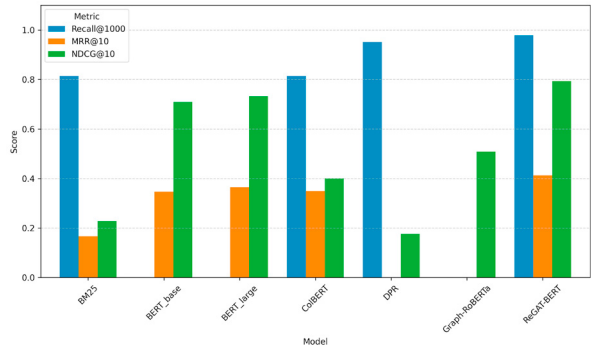


Fig. 2. Comparison of ReGAT-BERT with State-of-the-Art Baselines

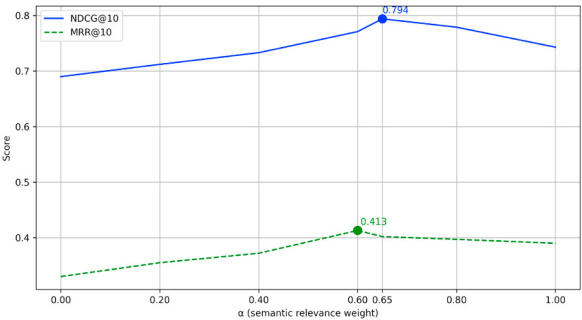


Fig. 3. Impact of α (semantic relevance weight) on NDCG@10 and MRR@10

These results validate the architectural choices of ReGAT-BERT, particularly the effectiveness of dynamic reranking. By using context-aware reranking, our model ensures that highly relevant passages are prioritized, addressing limitations in initial retrieval stages. The integration of semantic relevance and contextual similarity within the reranking process significantly enhances ranking accuracy, leading to more precise and coherent retrieval results.

5. Limitations of the proposed approach

Our approach which is based on the integration of transformer models and GATs for dynamic reranking tasks has shown promising results in capturing both sequential dependencies and structural relationships within data. However, several limitations hinder their full potential. First, the computational complexity of combining these architectures often leads to scalability issues, particularly when dealing with large-scale datasets. Second, the dynamic nature of reranking tasks requires real-time adaptability, which is challenging due to the static nature of pre-trained transformer models and the difficulty in updating graph structures. Third, the interpretability of the integrated model remains limited, making it difficult to understand how the combination of attention mechanisms influences reranking decisions. Additionally, the reliance on high-quality graph construction for GATs can introduce biases or errors if the graph representation is suboptimal. Finally, the integration often requires extensive hyperparameter tuning and domain-

specific customization, which can be resource-intensive. Addressing these limitations is crucial for advancing the applicability of transformer hybrid-GAT models in dynamic reranking scenarios.

6. Conclusion and prospects

In this study, we introduced a novel passage reranking method designed to enhance information retrieval performance by seamlessly integrating lexical-semantic relevance with contextual information. By using transformer models, specifically Cross-Encoder BERT and Bi-Encoder architectures enhanced through Graph Attention Networks (GAT), our approach effectively addresses limitations associated with traditional lexical matching and purely semantic-based retrieval systems. Our advanced experimental evaluations demonstrate that the proposed hybrid approach substantially improves retrieval accuracy, semantic precision, and contextual coherence compared to existing state-of-the-art methodologies. Moreover, by dynamically reranking passages based on combined semantic and contextual scores, our framework achieves superior performance while efficiently managing computational resources, overcoming typical scalability challenges. This research not only advances the theoretical understanding of effective passage retrieval but also offers practical solutions that can be readily adopted in real-world applications involving complex textual data processing.

In future work, we aim to explore two new directions. The first involves conducting a more in-depth study using data collections and analyses to address the challenge of dynamic reranking. Through this effort, we hope to validate and enhance the performance of our model. The second direction focuses on using machine learning techniques for dynamic reranking, which has the potential to substantially improve the effectiveness and outcomes of our approach.

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